

Solutions

1. This is Exercise 5.2.

2. Denote the above matrix by A . We have

$$\begin{aligned}
 \begin{pmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{pmatrix} &\xrightarrow{R_3 - 4R_1} \begin{pmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 0 & -3 & -6 \end{pmatrix} \\
 &\xrightarrow{R_2 - 2R_1} \begin{pmatrix} 2 & 3 & 4 \\ 1 & 0 & -1 \\ 0 & -3 & -6 \end{pmatrix} \\
 &\xrightarrow{R_2 \leftrightarrow R_1} \begin{pmatrix} 1 & 0 & -1 \\ 2 & 3 & 4 \\ 0 & -3 & -6 \end{pmatrix} \\
 &\xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 3 & 6 \\ 0 & -3 & -6 \end{pmatrix} \\
 &\xrightarrow{R_3 + R_2, \frac{1}{3}R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{C_3 + C_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{C_3 - 2C_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

so the rank of A is 2.

3. Note that $99 = 3^2 \cdot 11$. So by the Fundamental Theorem of Finite Abelian Groups, the isomorphism classes are

$$\mathbb{Z}_{99} \quad \text{and} \quad \mathbb{Z}_3 \oplus \mathbb{Z}_{33}$$

As $n_{11} \mid 3^2$, we have $n_{11} \in \{1, 3, 3^2\}$. So $n_{11} = 1$ and the Sylow 11-subgroup is normal. Likewise we deduce that $n_3 = 1$, so there is no non-abelian group of order 99.

4. The concept of a Jordan normal form was covered in the lectures this time, so one then has that A is conjugate to an upper triangular matrix with zero along the diagonal. As A^2 has the zeros also in the off-diagonal entries immediately above the above, and similarly for higher powers of A (i.e. one gets another diagonal line of zeros), we see that $A^n = 0$.
5. (a) Since the tensor product is distributive with respect to direct sums, we have that

$$\begin{aligned} (\mathbb{Z} \oplus \mathbb{Z}) \otimes_{\mathbb{Z}} ((\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}) \oplus (\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z})) &\cong (\mathbb{Z} \otimes_{\mathbb{Z}} (\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z})^2)^2 \\ &\cong ((\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z})^2 \otimes_{\mathbb{Z}} \mathbb{Z})^2 \\ &\cong (\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z})^4, \end{aligned}$$

where we have used Theorem 11.10(i) for the second isomorphism and Theorem 11.9 for the final isomorphism. So the answer is $\mathbb{Z}^{12}$.

(b) Arguing as in Exercise 15.1(ii), we have $\mathbb{Z}_2 \otimes_{\mathbb{Z}} \mathbb{Z}_6 \cong \mathbb{Z}_2$.

6. Either R has non-zero nilpotent ideals, then we are done, by choosing $0 \neq a \in R$ such that $a^n = 0$ and then we can take a and a^{n-1} are our two elements.

Or by the Wedderburn-Artin Theorem, we have that

$$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

for some $k \in \mathbb{N}$ with $n_1, \dots, n_k \in \mathbb{N}$ and $D_1, \dots, D_k$ division rings. Since R is not a division ring, it follows that either $k > 1$ or if $k = 1$ then $n_1 > 1$. Hence we can easily find matrices $A, B \neq 0$ such that $AB = 0$.